Classical Encryption Methods

The family of classical private key ciphers include shift ciphers, affine ciphers, and transposition ciphers. These types of ciphers involve a private key which is used to encrypt and decrypt the messages (M) and the ciphertext (C).

<u>Shift cipher:</u> $C = f(M) = (M + k) \bmod 26$, where $\{k\}$ is the private key.

<u>Affine cipher:</u> $C = f(M) = (aM + b) \bmod 26$, where $\{a, b\}$ is the private key.

<u>Transposition cipher:</u> M is split into blocks of size n, $M_1 M_2 ... M_n$, and a permutation of the set $\{1, 2, ..., n\}$ is the private key.

<u>Vigenere cipher:</u> M is split into blocks of size n, $M_1 M_2 ... M_n$, and a keyword with numerical equivalent $k_1 k_2 ... k_n$ is the private key such that $C_i = f(M_i) = (M_i + k_i) \bmod 26$.

Note: the shift, affine, and Vigenere ciphers use mod 26 and require that the plaintext messages are converted to integer strings using $\mathbb{Z}_{26}$. The transposition cipher can be used on original plaintext.

Examples: Encrypt "security" using:

1. The shift cipher $f(M) = (M + k) \bmod 26$, where $\{k = 10\}$:
Converting to an integer string: 18 04 02 20 17 08 19 24
Apply $f(M)$ to each 2-digit block: 02 14 12 04 01 18 03 08

2. The affine cipher $f(M) = (aM + b) \bmod 26$, where $\{a = 2, b = 3\}$:
Converting to an integer string: 18 04 02 20 17 08 19 24
Apply $f(M)$ to each 2-digit block: 13 11 07 17 11 19 15 25

3. The transposition cipher with permutation $\{3, 1, 2\}$.
Add a character _ and split into blocks of size $n = 3$: sec uri ty_
We can transpose the characters using the permuation: cse iur _ty

4. The Vigenere cipher with keyword "jazz":
Get numerical equivalent of "jazz": 09 00 25 25
Converting to an integer string: 18 04 02 20 17 08 19 24
Split into blocks of size $n = 4$: 18 04 02 20 17 08 19 24
Apply $(M_i + k_i) \bmod 26$: 01 04 01 19 00 08 18 23

Exercises:

1. Encrypt "sheridan" using:

a) The shift cipher $f(M) = (M + k) \bmod 26$, where $\{k = -5\}$

b) The affine cipher $f(M) = (aM + b) \bmod 26$, where $\{a = 7, b = 12\}$

c) The transposition cipher with permutation $\{3,1,4,2\}$

d) The Vigenere cipher with keyword "dual"

2. What is the decryption function, $f^{-1}(M)$, for the affine cipher $f(M) = (15M - 9) \bmod 26$?

3. What is the decryption permutation for the transposition cipher with permutation $\{2,5,1,3,4\}$?

4. What is the decryption keyword for the Vigenere cipher with keyword "end"?

Answers:

1.

a) 13 02 25 12 03 24 21 08
b) 08 09 14 01 16 07 12 25
c) esrhaind
d) 21 01 04 02 11 23 00 24

2. $f^{-1}(C) = (7C + 11) \bmod 26$

3. $\{3,1,4,5,2\}$

4. “wnx”